RCS01 Student Guide

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big Idea

Residential construction is not just hammering boards together. A building is a set of connected systems. Good tradespeople understand the system, work safely, verify what they did, and leave the work ready for the next person.

Major Residential Systems

System	Plain Meaning	Example
Structure	What holds the building up	Studs, plates, headers, sheathing
Electrical	How power is routed and controlled	Receptacles, switches, lights, breakers
Plumbing	How water enters and waste leaves	Supply lines, drains, traps, shutoffs
HVAC	How air and heat move	Vents, ducting, filters, heat pumps
Envelope	How the building separates inside from outside	Windows, insulation, air sealing
Finish	What protects and completes surfaces	Drywall, trim, paint, flooring

Professional Safety Habits

- Wear PPE that matches the task.
- Ask before using unfamiliar tools.
- Check the work area before starting.
- Use stop-work authority when unsure.
- Tell the truth about mistakes early.
- Clean as you go.

Power Tool Introduction

- Drill/driver: holes, screws, and fasteners. Main hazards: bit slips, stripped fasteners, pinched fingers.
- Circular saw: straight cuts in sheet goods and lumber. Main hazards: blade path, kickback, support, cord path.
- Miter saw: controlled crosscuts. Main hazards: hand placement, blade coast-down, material support.
- Jigsaw/oscillating tool: cutouts and detail cuts. Main hazards: wandering blade, hidden material, hand placement.

Rolling Sink Lab Purpose

Today's lab teaches systems thinking. You will assemble a simple plumbing drain setup, test it, find leaks if they happen, correct them, and document what you verified.

Today's Field Note

A professional is not the person who never makes a mistake. A professional is the person who notices early, stops safely, fixes cleanly, and leaves evidence that the work was checked.